

S.H.A.D.E.

Synthetic Host for Automated Data Extractor

Target Event: KPRIET 24-Hour Hackathon | Domain Track: AI & Tech | Team Structure: 4 B.Tech Undergraduates | Core Architecture: Autonomous AI Sentinel & Real-Time DLP

1. Exact Problem Statement

Core Problem Statement (4 Lines):

In the era of cloud AI and pervasive web surveillance, individuals and organizations continuously suffer irreversible data exfiltration from unmonitored human copy-pasting of confidential credentials and Indian PII into third-party servers, while commercial brokers aggressively compile unauthorized shadow profiles. Although statutory privacy frameworks like the Indian DPDP Act 2023 grant citizens the legal right to data erasure, ordinary users have zero automated tooling to intercept leaks in real time or enforce statutory takedowns.

2. The Solution & How We Are Solving It

S.H.A.D.E. shifts cybersecurity from passive defense to an active **Personal Autonomous Counter-Intelligence & Security Operations Center (SOC)**. It acts as a local Synthetic Host that manages, audits, cloaks, and enforces statutory data rights across a 360-degree lifecycle:

Lifecycle Phase	Engine / Mechanism	How It Solves The Problem
1. PREVENT (Real-Time DLP)	Deterministic Regex + Verhoeff Checksum + Synthetic Tokenizer	Monitors outbound prompts and clipboard in real time. Before Aadhaar, PAN, API keys, or passwords leave the host to cloud AI, they are intercepted and replaced with safe synthetic placeholders (e.g., <SYN_AADHAAR_9921>). Plaintext never touches external servers.
2. DETECT (Data Extractor)	OSINT Breach Correlation & Broker Scraper Swarm	Autonomously correlates user identities against public breach dumps (HaveIBeenPwned API) and data broker registries to calculate an algorithmic 0–100 Exposome Threat Index and map exposed digital nodes.
3. CLOAK (Synthetic Host)	Decoy Persona Engine & Canary Honey-Tokens	Generates trackable synthetic decoy credentials for web registrations. If an external service leaks or sells the user's data, the canary token immediately trips an alert, providing cryptographic leak attribution.
4. ENFORCE (Statutory Takedown)	DPDP Act 2023 Section 12 Auto-Notice Generator	Leverages LLM reasoning (Gemini Flash / Local Ollama) to extract Data Fiduciary endpoints and autonomously draft formal statutory erasure requisitions with verified legal clauses, enabling 1-click legal enforcement.
5. INTERACTION (Ambient Voice)	Offline 'Hey Shade' Porcupine Engine + Vosk STT	Allows hands-free voice commands ('Hey Shade, scan footprint', 'Hey Shade, deploy cloaking') with spoken TTS audio briefings and a live Cyber War-Room HUD.

3. Technical Requirements: What We Are Gonna Need

To build and demonstrate this prototype reliably within the 24-hour hackathon window, the project utilizes a lightweight, modern, and decoupled architecture:

Category	Exact Technology / Tool	Role & Justification in 24-Hour Sprint
Backend Server	Python 3.10+ & FastAPI / Uvicorn	High-performance asynchronous API layer; native Python integration for regex heuristics and LLM orchestrators.
DLP Heuristics	Verhoeff Algorithm & Indian Regex Suite	Mathematically verifies 12-digit Indian Aadhaar checksums to prevent false positives; catches PAN cards and secret API keys with zero latency.
AI Reasoning	Google Gemini 2.5 Flash / Local Ollama (Phi-3)	Drafts structured legal DPDP Section 12 notices and summarizes privacy policies in sub-second inference speeds.
War-Room UI	React 19 / Vite + Tailwind CSS + Lucide Icons	Cyberpunk SOC interface: real-time 0–100 threat gauge, live interception sandbox, footprint exposure graph, and mobile alert card.
Voice Daemon	Picovoice Porcupine (Hey Shade) + Vosk + Pyttsx3	Pre-configured local wake-word listener and offline speech-to-text; connects via REST to FastAPI for hands-free operations.
APIs & Data	HavelBeenPwned API & Curated Broker Registry	Real-world breach lookup paired with local cached fallback datasets for 100% offline demo resilience.

Team Division of Labor (4 B.Tech Undergraduates):

- **Member 1 (Backend & DLP):** FastAPI service, Aadhaar Verhoeff validation, regex secret scanners, synthetic tokenization.
- **Member 2 (AI & Legal Swarm):** Gemini prompt pipelines, DPDP statutory notice generators, HavelBeenPwned API integration.
- **Member 3 (Frontend & UX):** React/Vite dashboard, SVG Threat Index meter, live interception interactive sandbox.
- **Member 4 (Voice & Pitch Lead):** 'Hey Shade' audio integration, Wi-Fi failover cache, pitch deck & live presentation.

4. Technical Feasibility & 24-Hour Hackathon Execution

A common pitfall in hackathons is over-engineering. S.H.A.D.E is engineered with **strict hackathon pragmatism** to guarantee 100% completion:

- **Manageable Codebase Volume:** The entire project consists of ~1,000 lines of clean code across decoupled modules, allowing each student to write ~250 lines without merge conflicts.
- **Unfair Starting Advantage:** The voice wake-word engine (Porcupine 'Hey Shade' + Vosk) is already functional, eliminating 6–8 hours of risky audio driver debugging.
- **Zero Expensive Cloud Infrastructure:** Over 70% of processing (DLP regex, Verhoeff checksums, voice detection) runs locally on the CPU; no dedicated GPUs or CUDA drivers required.
- **The College Wi-Fi Shield (Zero-Crash Guarantee):** Incorporates a Dual-Engine toggle: attempts live cloud APIs first, but falls back instantly to local high-fidelity mock datasets if venue Wi-Fi drops.
- **Auditorium Noise Mitigation:** Includes an on-screen 'Voice Simulation HUD' with 1-click trigger chips so the demo runs seamlessly even in deafeningly loud presentation halls.

5. Business Model, Commercial Viability & Financial Returns

S.H.A.D.E is not merely a technical showcase—it is designed as a commercially viable, high-margin cybersecurity business:

Market Catalysts:

- 1. Statutory Compliance:** The Indian **DPDP Act 2023** imposes statutory penalties of up to **₹250 Crore (~\$30M USD)** for personal data breaches and non-compliance with user erasure requests.
- 2. The Shadow-AI Wave:** Over **85% of corporate employees** paste sensitive corporate/client data into cloud AI tools without authorization, creating massive enterprise liability.

Revenue Stream	Target Audience	Pricing & Commercial Model	Strategic Value
B2C SaaS (Freemium)	Privacy-conscious individuals, professionals, students	- Free: 10 scans/mo + 1 DPDP notice - Pro: ₹299/mo (₹2,499/yr) unlimited DLP & automated takedowns	Viral consumer adoption and low-friction user acquisition engine.
B2B Enterprise (DLP Shield)	Indian FinTechs, HealthTechs, IT consultancies, SMEs	₹499 / employee / month - Endpoints firewall against unauthorized prompt leakage into LLMs	High ARR enterprise contracts driven by fear of ₹250 Cr DPDP penalties.
Takedown API (B2B2C)	VPN providers, antivirus software, password managers	₹25 per verified erasure dispatch via developer API integration	Scalable zero-acquisition-cost affiliate distribution channel.

Metric	Value	Strategic Explanation
Average Revenue Per User (B2C)	₹299 / month	Competitive pricing compared to US alternatives (\$10–\$12/month).
Cost of Goods Sold (COGS)	~₹25 / user / month	Edge-first architecture keeps cloud API calls minimal (local regex/Verhoeff = 0 compute).
Gross Profit Margin	91.6%	Matches elite tier-1 SaaS business profitability standards.
Initial Seed Capital Needed	< ₹30,000	Covers cloud domain, SSL, VPS hosting, and baseline LLM credits for first 1,000 users.
Year 1 Financial Projection	₹1.3 Crore ARR	Based on 2,500 B2C Pro subscribers + 8 pilot SME clients (1,200 seats). Cashflow positive by Month 4.
Year 2 Financial Projection	₹6.2 Crore ARR	Expansion across Indian corporate sectors and regional DPDP enforcement compliance.

Judges' Takeaway Summary:

S.H.A.D.E solves an authentic, high-urgency problem at the intersection of AI, Indian cybersecurity, and privacy legislation. It combines demonstrable engineering depth (DLP tokenization, OSINT breach graphs, voice wake-word) with an indisputable 91% gross-margin business model. It is 100% buildable, defensible, and testable by 4 B.Tech students in 24 hours.